

TIME SERIES ANALYSES

Advanced Drug Utilization Research - ISPE 2026

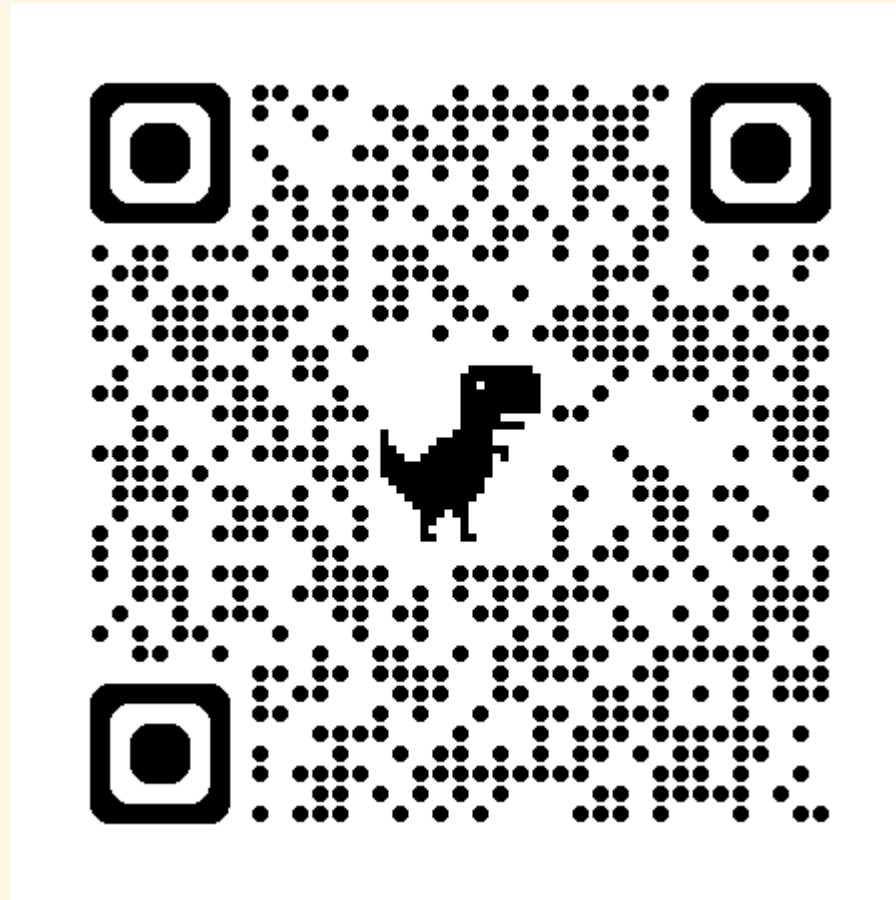
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Slides & script

<https://github.com/martinkhho/ispe-time-series-2026>



What you will learn

By the end of this session, you should be able to:

- Explain what is a **time series**
- Explain how **time series analyses** can be used in drug utilization research
- Use **regression** and **ARIMA** to model time series
- Conduct **interrupted time series analyses**

Disclosures

- I have no personal or financial relationships relevant to this presentation
- Some slides were adapted from last year's talk by Dr. Mina Tadrous

Declaration of AI-assisted technologies

All instructional material in this presentation was written by the author except where external sources are cited. OpenAI Codex (GPT-5.6-Sol, reasoning: ultra) helped format the slide deck. The author reviewed, edited, and verified the formatting suggestions as needed and takes full responsibility for the final product.

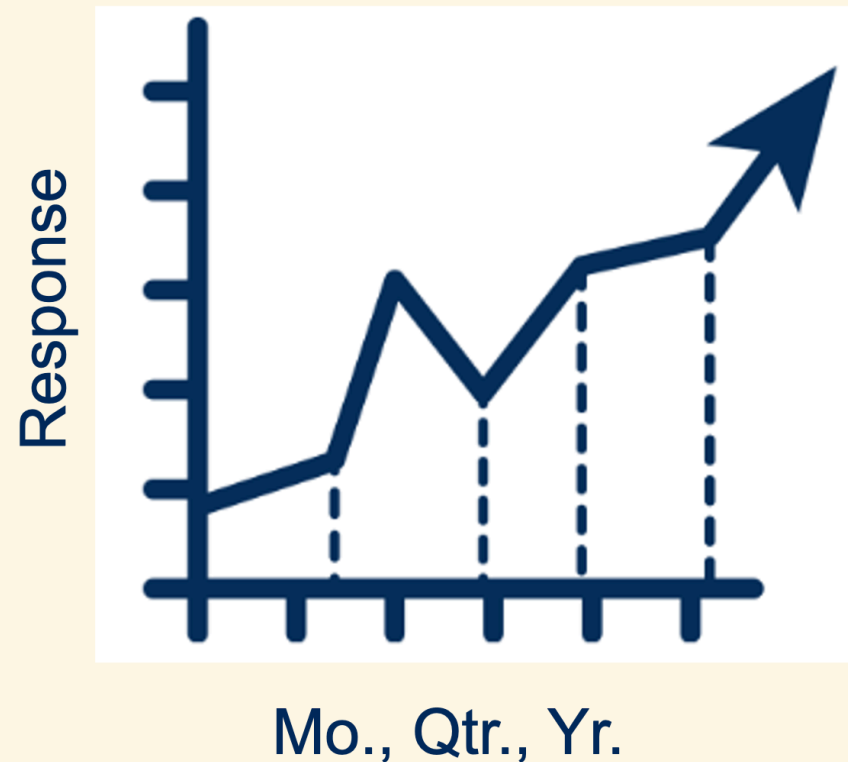
Univariate time series

- A set of observations of the same variable, indexed by time
- Usually, at regular time intervals
- **Key difference from usual data:** observations are not necessarily independent

$$\begin{aligned}x_t &= x_{t-1} + w_t \\ &= (x_{t-2} + w_{t-1}) + w_t\end{aligned}$$

Time series analyses

- Used across many fields, most commonly in finance and economics
- From pharmacoepidemiology lens: a repeated cross-sectional analysis



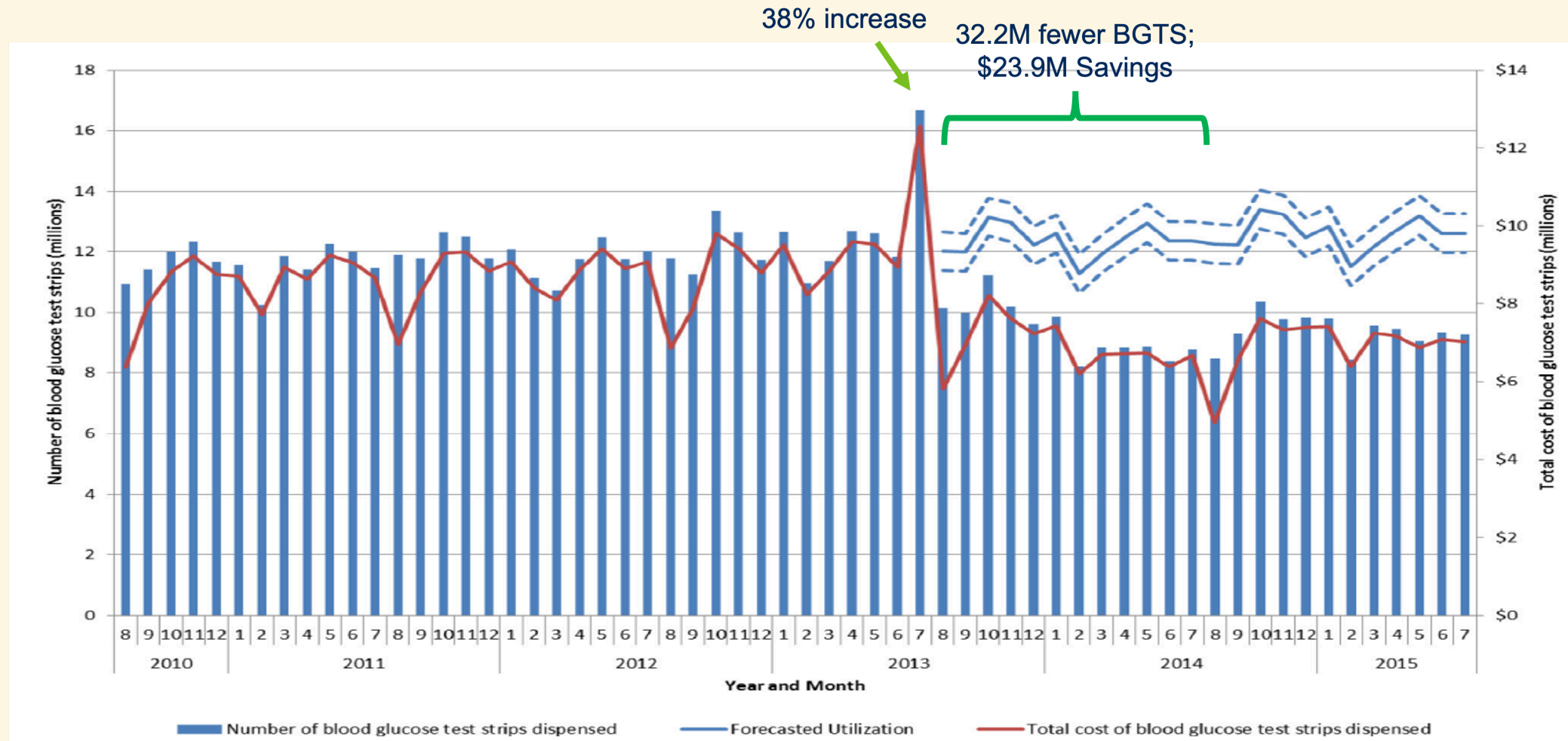
Motivating example¹ - background

- The province of Ontario, Canada, introduced limits on blood glucose test strips (BGTS) based on diabetes therapy in 2013

Diabetes group	Test strip limits
Insulin users	Max 3,000/year
Hypoglycemia-inducing oral antihyperglycemics	Max 400/year (~30/month)
All others	Max 200/year (~16/month)

- Concern over the impact of this policy on:
 - Test strip utilization and spending
 - Clinical outcomes

Motivating example¹ - BGTS utilization & costs



BGTS utilization and costs in Ontario

Motivating example¹ - clinical outcomes

Design

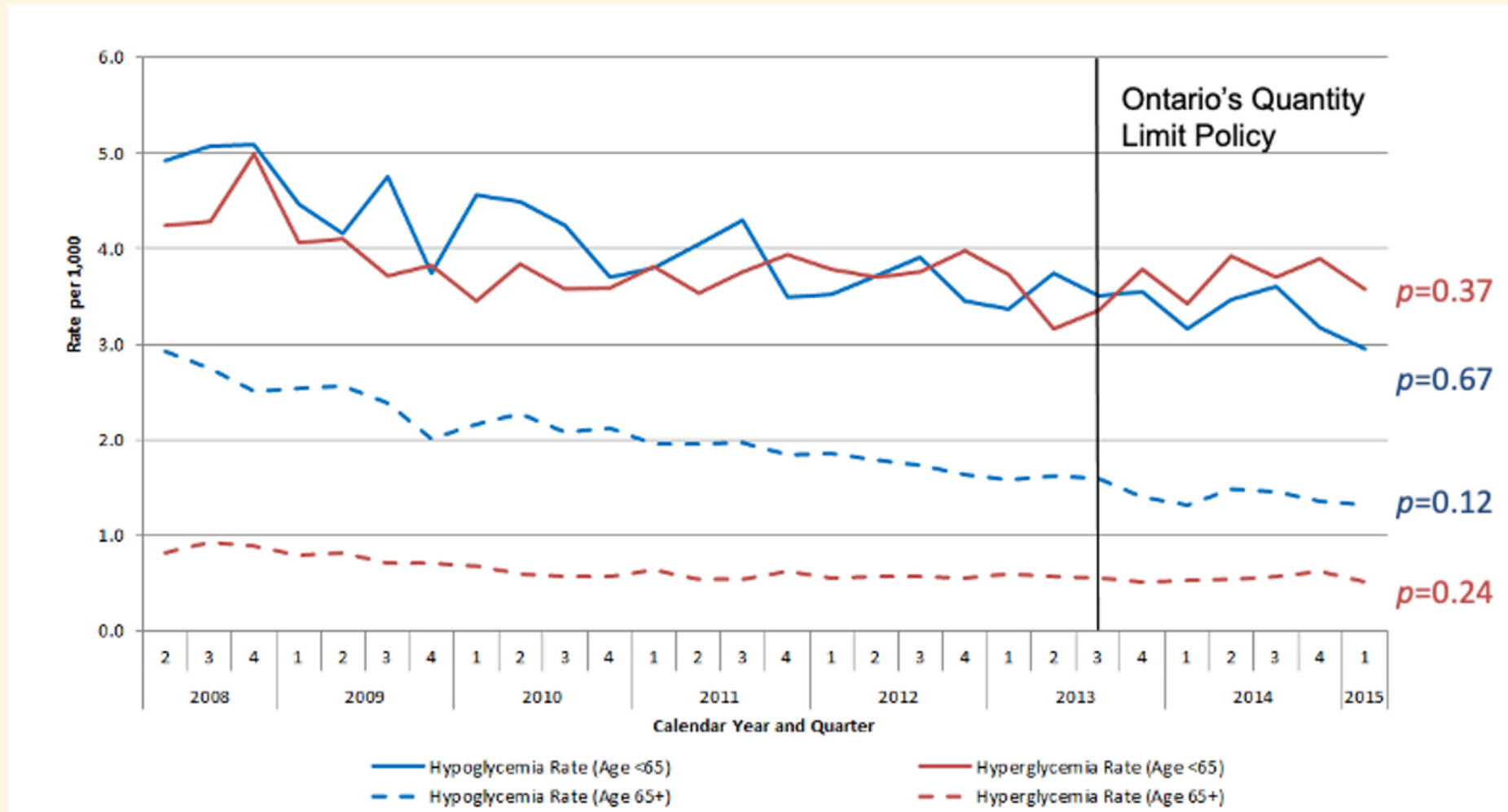
- Repeated cross-sectional analysis
- April 1, 2008 - March 31, 2015 (quarterly)
- Using interventional ARIMA models
- Stratified by age: <65 vs. 65+ years

Definitions

Users were hierarchically defined based on their diabetes drug therapy in the prior year to the end of the quarter:

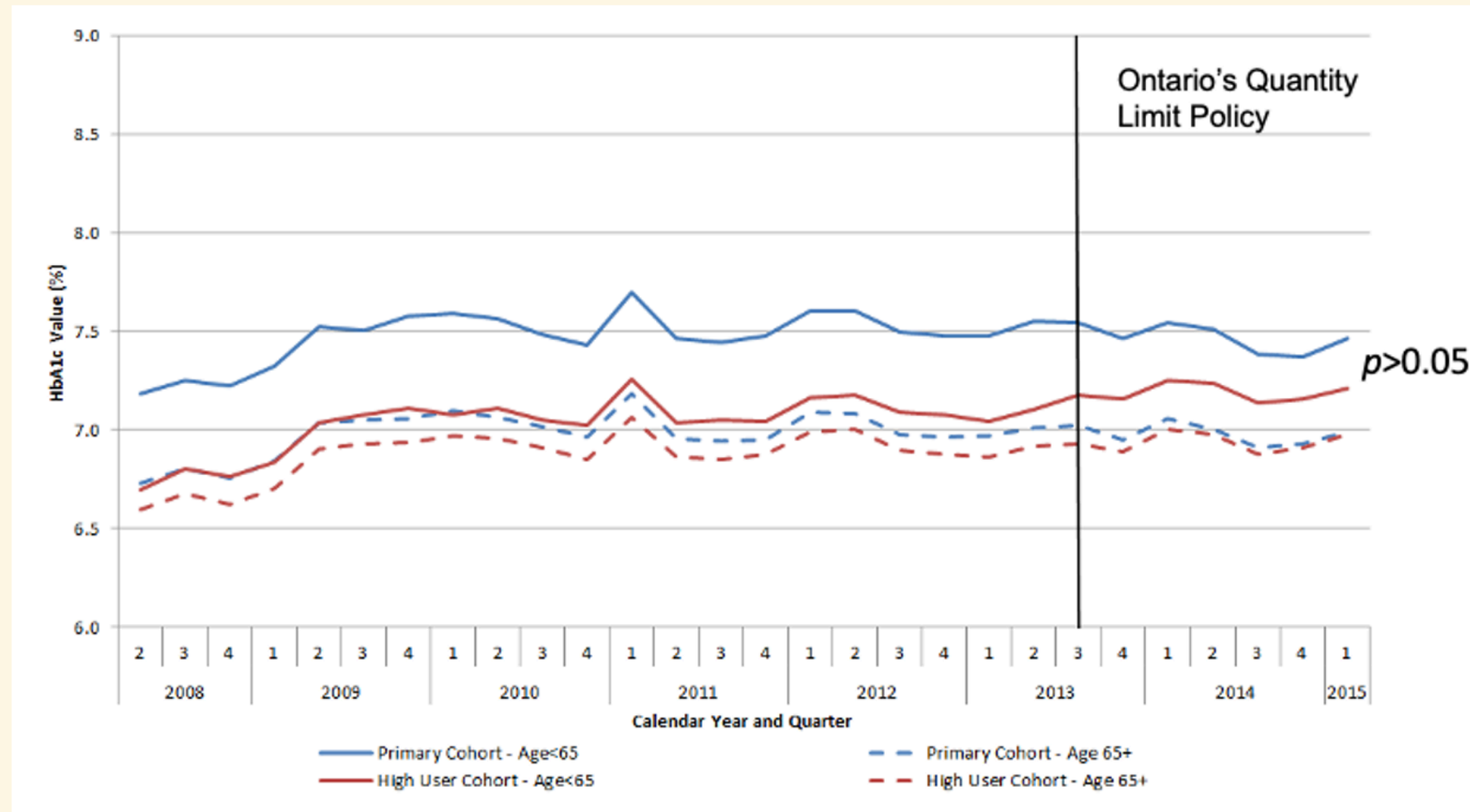
1. Insulin
2. Hypoglycemia-inducing oral antihyperglycemic
3. Other oral antihyperglycemic
4. No drug therapy

Motivating example - emergency department visits



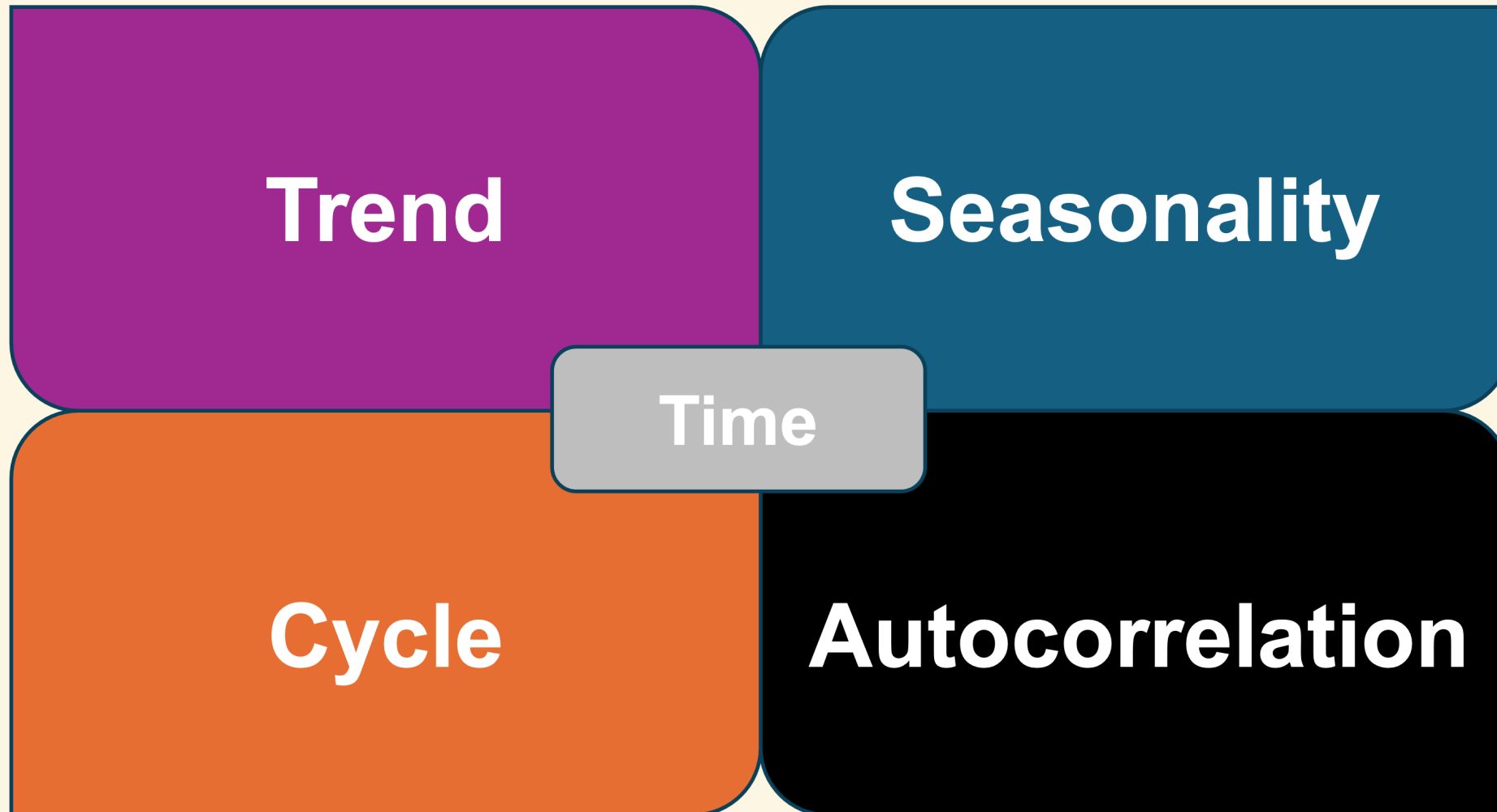
Rates of emergency department visits for hypoglycemia and hyperglycemia

Motivating example - mean HbA1c



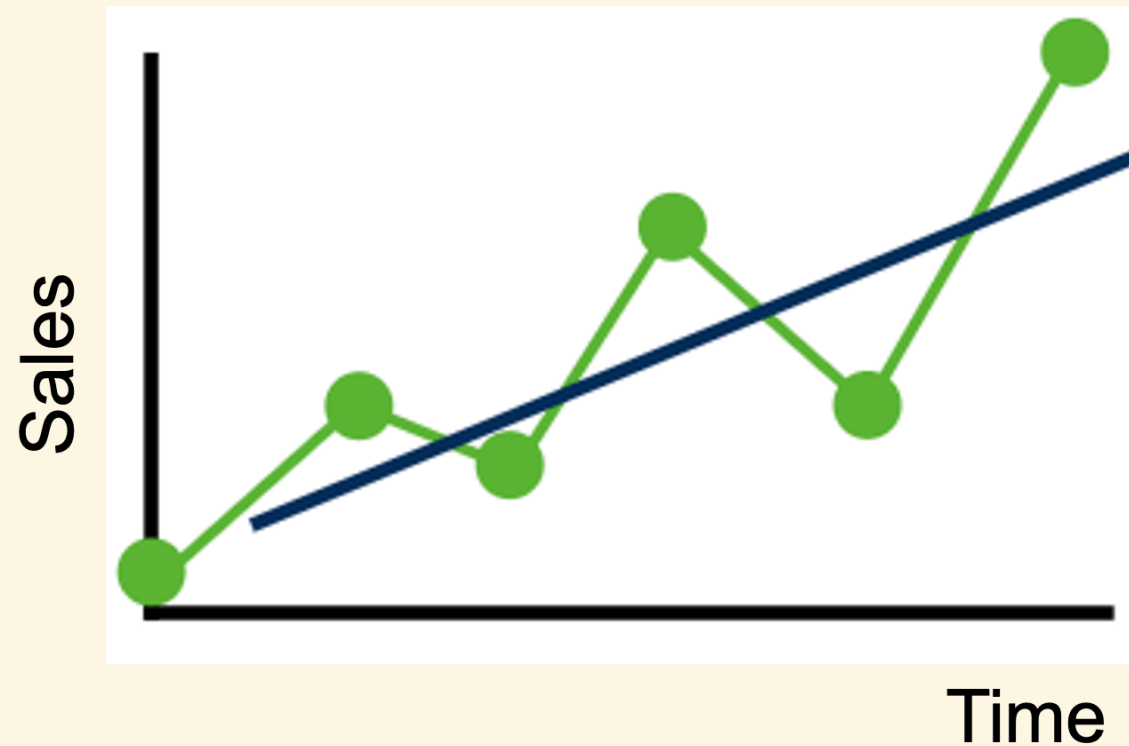
Mean HbA1c (overall and among high-use cohort)

Time series characteristics



Trend

- A persistent, overall upward or downward pattern
- May be due to population growth, shifting prescribing patterns, etc.



Seasonality

- Regular up and down fluctuations
- May be due to weather, prescription refills, etc.
- Occurs within one year (unlikely to have seasonality if you have annual data)

Cycles

- Upward and downward swings
- E.g., housing sales, immigration patterns
- Unlike seasonality, may vary in length, often lasting years on end

Autocorrelation

- The correlation between a data point and a lagged version of itself
- Can be positive or negative
- E.g., stock market, “hot/cold streaks” in sports



Other

Interventions

- Policy/formulary change
- Supply chain disruption
- Pandemic

Irregularities/randomness

- Not systematic
- Residual fluctuations
- Unrepeating

Time series analyses

What kinds of questions do we ask?

- Impact of interventions
- Forecasting
- Descriptive analysis
- Trend analysis

Time series analyses

Approaches

- Regression models that regress x_t on t
- ARIMA models that relate the present value to a series of past values and shocks

In ARIMA, the idea is to iteratively model the time series process – you'll know when you're successful because the residuals will be patternless **white noise**:

- Mean 0
- Variance constant
- Uncorrelated

Regression

$$x_t = f(t)$$

- **Trend:** e.g., linear t , quadratic t^2
- **Seasonality:** an indicator variable for each season

For example, a time series with **linear trend** and **quarterly seasonality** may be modelled with:

$$x_t = \beta_1 t + \alpha_1 S_1 + \alpha_2 S_2 + \alpha_3 S_3 + \alpha_4 S_4 + \epsilon_t$$

ARIMA

- **AR:** autoregression
- **I:** integrated
- **MA:** moving average

Autoregression (AR model)

AR(1) means that we use the last 1 observation to predict the present time:

$$\begin{aligned}x_t &= \delta + \phi x_{t-1} + w_t \\&= \delta + \phi(\delta + \phi x_{t-2} + w_{t-1}) + w_t \\&\dots\end{aligned}$$

- δ is the intercept
- ϕ is the slope (persistence); $|\phi| < 1$ for stationary series (more about this later), meaning the influence of previous lags decays over time
- $w_t \stackrel{iid}{\sim} N(0, \sigma_w^2)$

Autoregression (AR model)

t	x_t	x_{t-1}
1	8	*
2	10	8
3	11	10
4	7	11
5	9	7

- The present value carries information from everything before, because the relationship is transmitted through intermediate lags

Autoregression (AR model)

- AR(1)

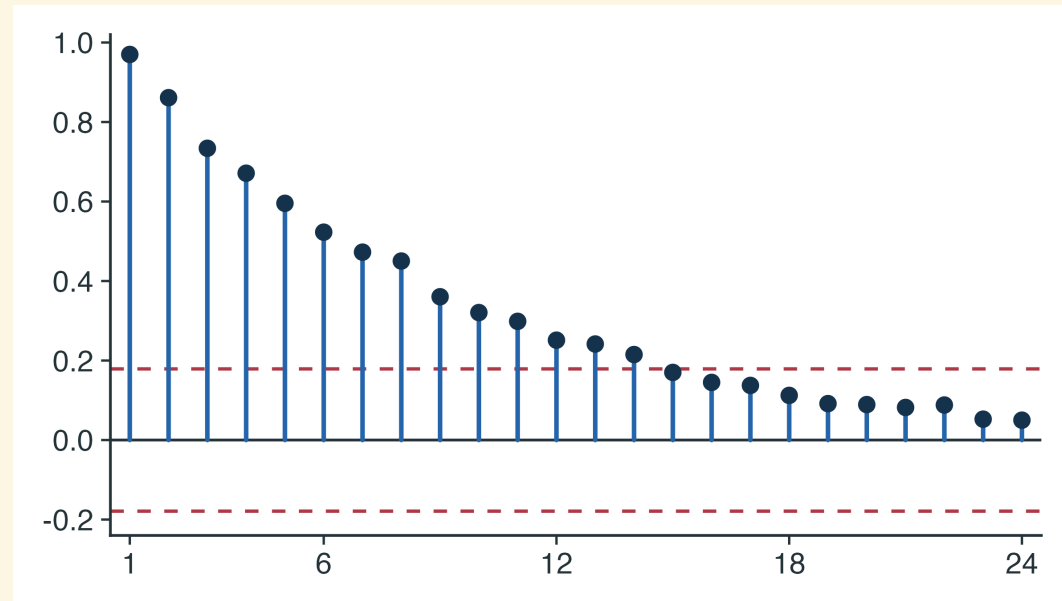
$$x_t = \delta + \phi_1 x_{t-1} + w_t$$

- AR(2)

$$x_t = \delta + \phi_1 x_{t-1} + \phi_2 x_{t-2} + w_t$$

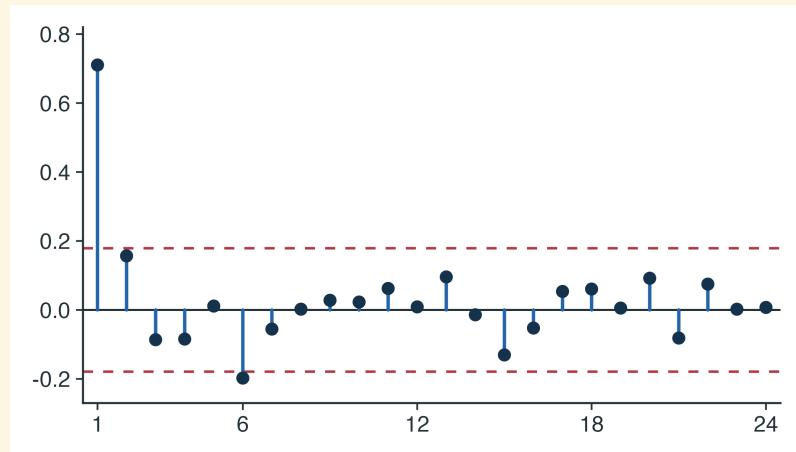
Autocorrelation function (ACF)

- Correlations between x_t and lagged values ($t - 1, t - 2, t - 3, \dots$)
- Useful for applying to both the raw series, as well as the model residuals



Partial autocorrelation function (PACF)

- Correlation between x_t and x_{t-h} after conditioning on the values in between
- Does x_{t-h} have a direct effect on x_t that is not via intermediate values?



- Very useful for determining AR orders
- PACF will spike at the corresponding lag, then shut off to zero afterwards

Stationarity

- A series is (weakly) stationary if:
 - Mean constant
 - Variance constant
 - Covariance between x_t and x_{t-h} is the same for all t at each lag h
- White noise is a special type of stationarity
- Stationarity may be addressed with **differencing**

ACF and PACF patterns (later) will not have their standard interpretations until the series is stationary. ACF of a non-stationary series will decay very slowly.

Stationarity

Trend violates stationarity

Augmented Dickey-Fuller (ADF) test:

$p > 0.05$ suggests a need to **difference**.

Seasonality violates stationarity

Osborn-Chui-Smith-Birchenhall (OCSB) test:

$p > 0.05$ suggests a need to **seasonally difference**.

Differencing

- Think of it as differentiation but for a discrete variable
- First differencing:

$$\nabla x_t = x_t - x_{t-1}$$

- Second differencing:

$$\nabla^2 x_t = \nabla x_t - \nabla x_{t-1}$$

- We often need first differencing, and rarely more than second differencing
- Over-differencing can introduce autocorrelation where none existed before

Seasonal differencing

- Seasonal (here, monthly) differencing:

$$\nabla_{12} x_t = x_t - x_{t-12}$$

- Seasonality skips intermediate lags
- Not the same as twelfth differencing

Moving average (MA model)

AR processes are not the only way to generate autocorrelation

- AR captures autocorrelation in past *values* (e.g., x_{t-1})
- MA captures autocorrelation in past *shocks* (e.g., w_{t-1})

Shocks are prediction errors

- In AR, the present value carries information from everything before
- In MA, the present value carries information from a fixed number of lags

Moving average (MA model)

- MA(1)

$$x_t = \mu + \theta_1 w_{t-1} + w_t$$

- MA(2)

$$x_t = \mu + \theta_1 w_{t-1} + \theta_2 w_{t-2} + w_t$$

ARIMA

- Can include AR terms, differencing operations, and MA terms
- Technically, “IARMA” in the sense that you should difference (if needed) first, to ensure stationarity, before figuring out AR and MA terms
- Without seasonality: $ARIMA(p, d, q)$
- With seasonality: $ARIMA(p, d, q) \times (P, D, Q)_S$

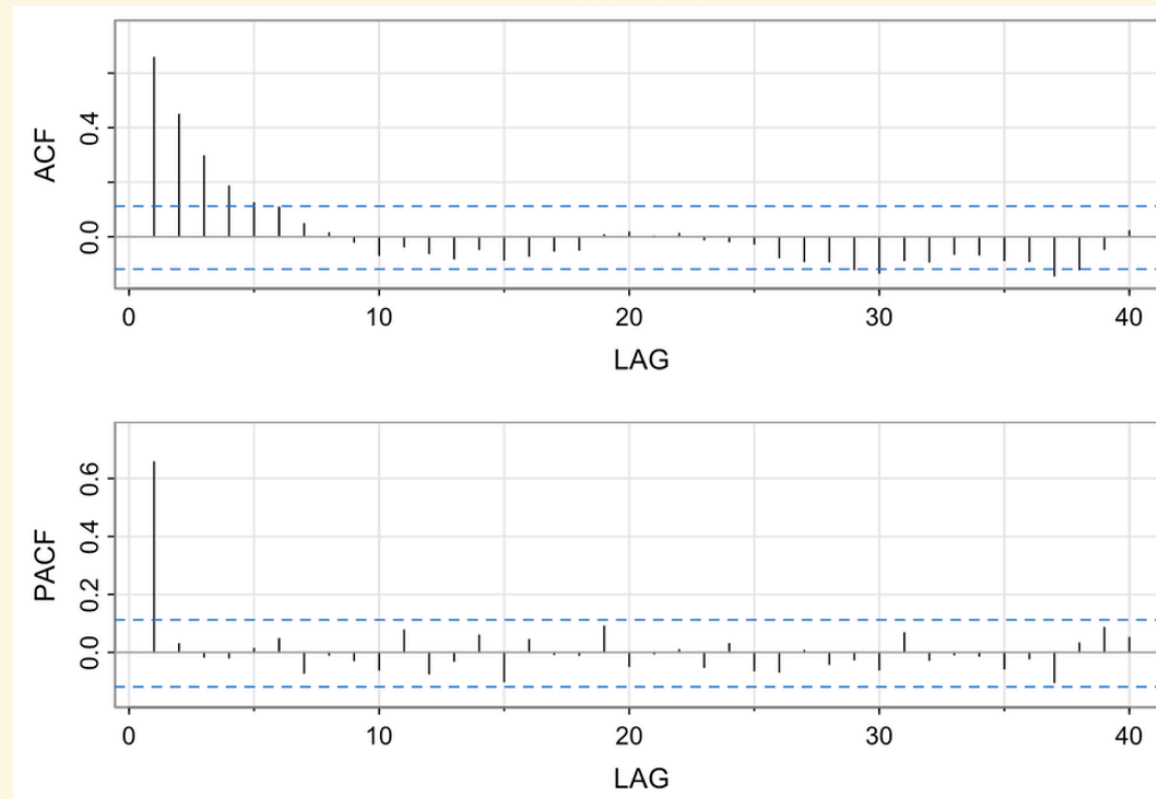
Correlogram patterns

True model	ACF	PACF
AR	Tapers	Cuts off past p
MA	Cuts off past q	Tapers
ARMA	Tapers	Tapers

- May not see clean patterns in sample data
- Sign (+/-) of autocorrelation does not matter for identifying AR and MA orders
- If all lags are non-significant, suggests that series is white noise

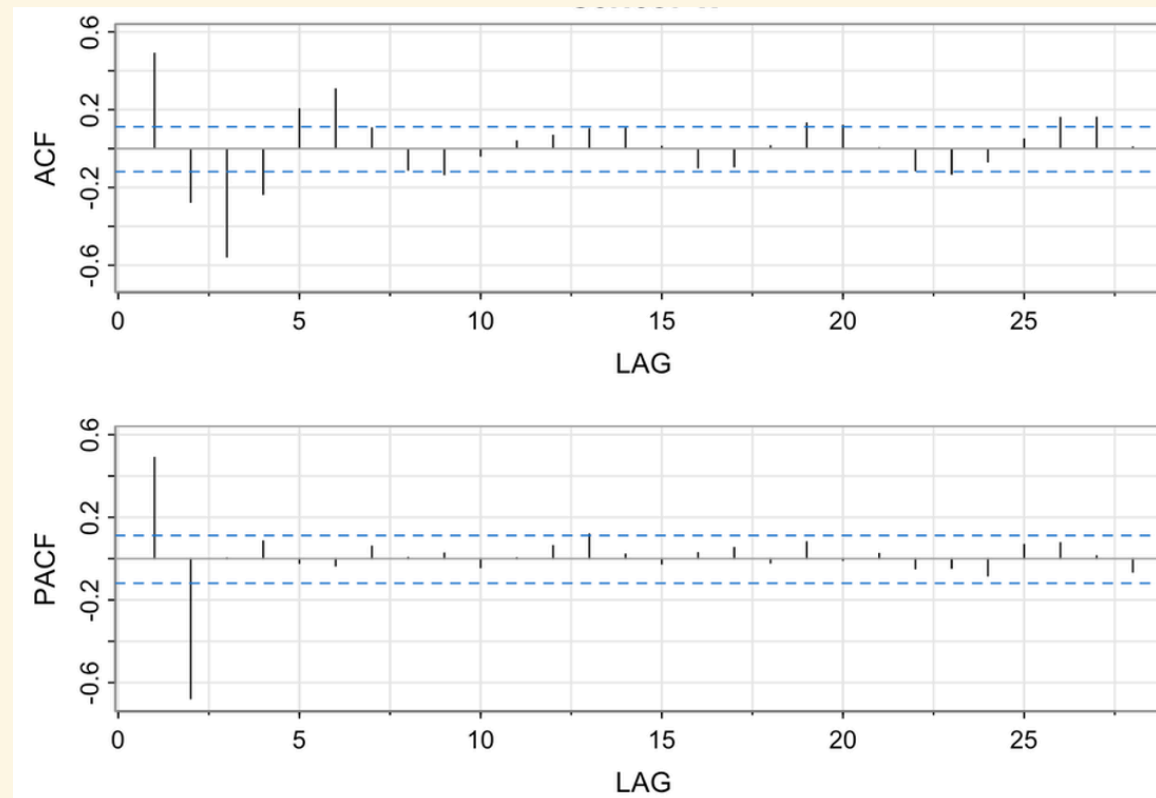
Identify the model:² #1

True model	ACF	PACF
AR	Tapers	Cuts off past p
MA	Cuts off past q	Tapers



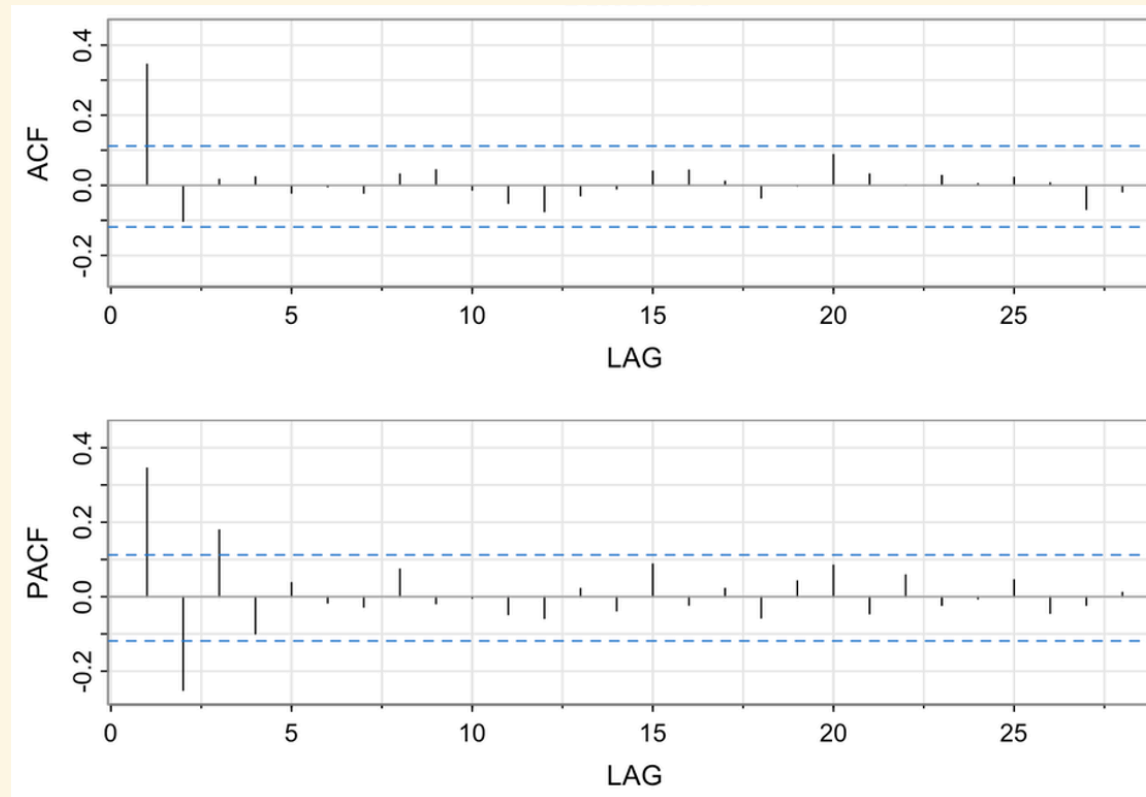
Identify the model:² #2

True model	ACF	PACF
AR	Tapers	Cuts off past p
MA	Cuts off past q	Tapers



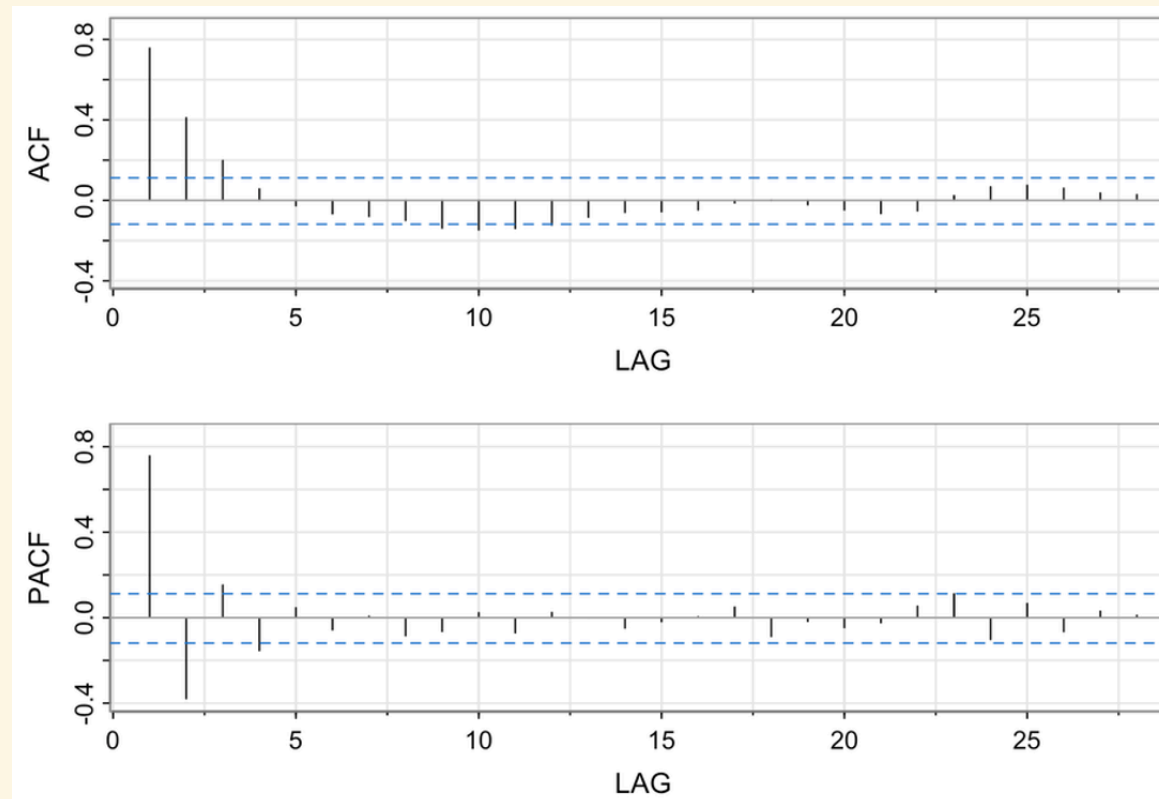
Identify the model:² #3

True model	ACF	PACF
AR	Tapers	Cuts off past p
MA	Cuts off past q	Tapers



Identify the model:² #4

True model	ACF	PACF
AR	Tapers	Cuts off past p
MA	Cuts off past q	Tapers



Diagnostics

Check:	What do we want to see?
Coefficients	Significant
ACF plot of the residuals	Not significant
Ljung-Box test for white noise of the residuals	Not significant
Q-Q plot of the residuals	Normality
Histogram of the residuals	Normality
Residuals-versus-fits plot	No heteroskedasticity
Time series plot of the residuals	No heteroskedasticity
Forecasts	Reasonable (e.g., not decreasing when expect to increase)

- Try a different ARIMA model if unsatisfactory

Comparing models

- Parsimony (low orders)
- Smallest standard errors of forecast values
- Best model fit, while not being overly complex – smaller values of:
 - Akaike information criterion (AIC)
 - AIC with correction for small sample size (AICc)
 - Bayesian information criterion (BIC)

ARIMA steps

1. Plot the time series

- Pay attention to seasonality and trends, informed also by OCSB and ADF tests
- If both seem possible, do seasonal differencing first as seasonality tends to dominate, then re-evaluate
- If heteroskedasticity, consider log transform

ARIMA steps

2. Plot correlograms

- Examine the ACF and PACF
- Glean AR and MA terms at low lags, and seasonal AR and MA terms at lags at multiples of S

3. Estimate the model

4. Perform diagnostics and select the best model

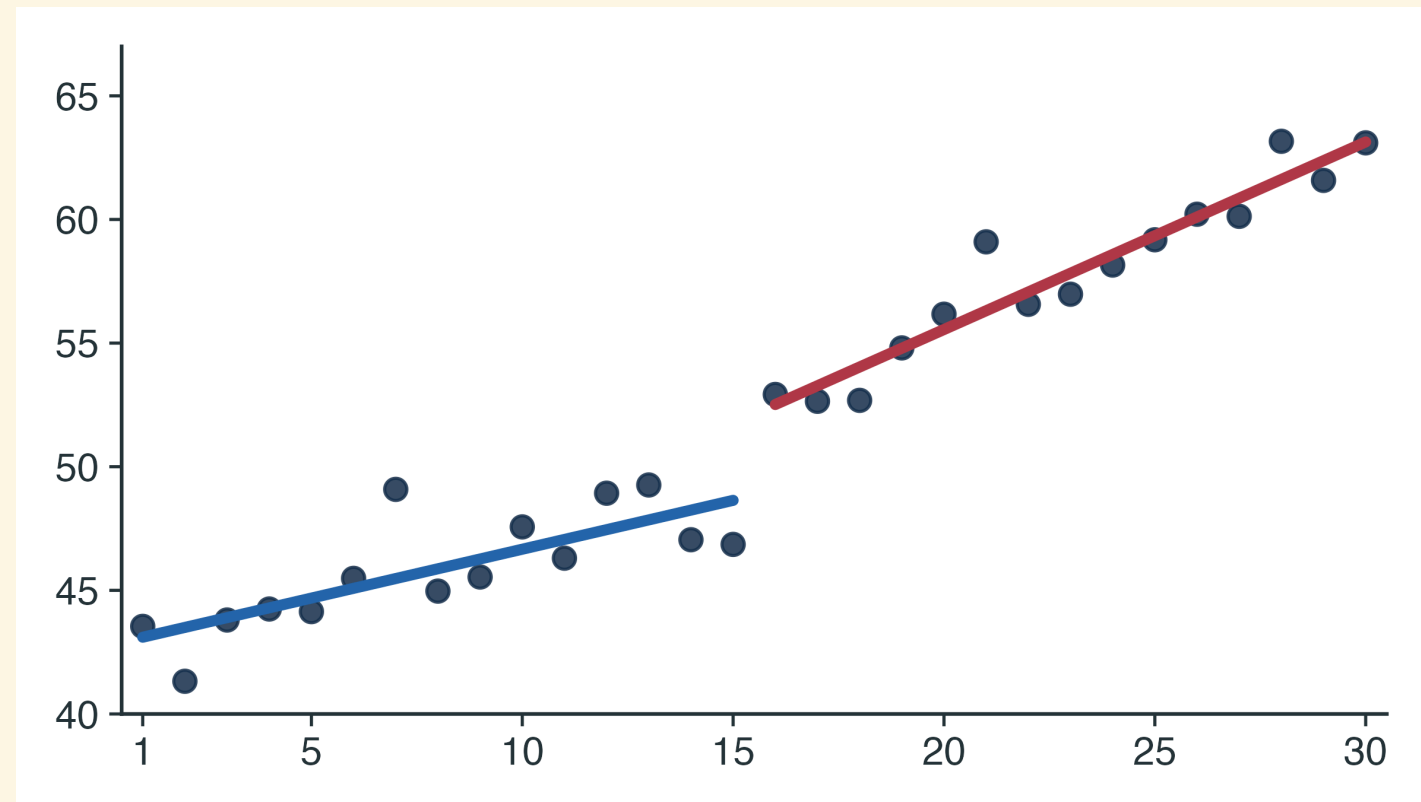
- Iterate if necessary

Intervention analysis

- Goes beyond modelling the time series; also estimate the effect of an intervention (e.g., policy, formulary change, supply chain disruption, pandemic)
- Ecological; in Ontario, Canada, we use administrative claims data stored at ICES
- **Quasi-experimental studies (QES):** intervention is as good as random, because assignment is plausibly unrelated to the outcome
- **Interrupted time series (ITS) analysis** is one type of QES; counterfactual is what the time series would have been had the intervention not occurred
 - Segmented regression
 - Interventional ARIMA

Intervention analysis: segmented regression

$$y_t = \beta_0 + \beta_1 t + \beta_2 I_t + \beta_3 (t \times I_t) + \epsilon_t$$



Intervention analysis: segmented regression

Let's break this down into two regression lines (set $t = 0$ for time of intervention):

Before intervention ($I_t = 0$) : $y_t = \beta_0 + \beta_1 t + \epsilon_t$

After intervention ($I_t = 1$) : $y_t = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)t$

- β_0 is the pre-intervention level (extrapolated) at the time of intervention
- β_1 is the pre-intervention slope (trend)
- β_2 is the immediate change in level at the intervention
- β_3 is the change in slope (trend) following the intervention
- $\beta_1 + \beta_3$ is the post-intervention slope

Intervention analysis: segmented regression

- Can model non-linear deterministic trend with polynomials or splines
- Can model deterministic seasonality using indicator variables

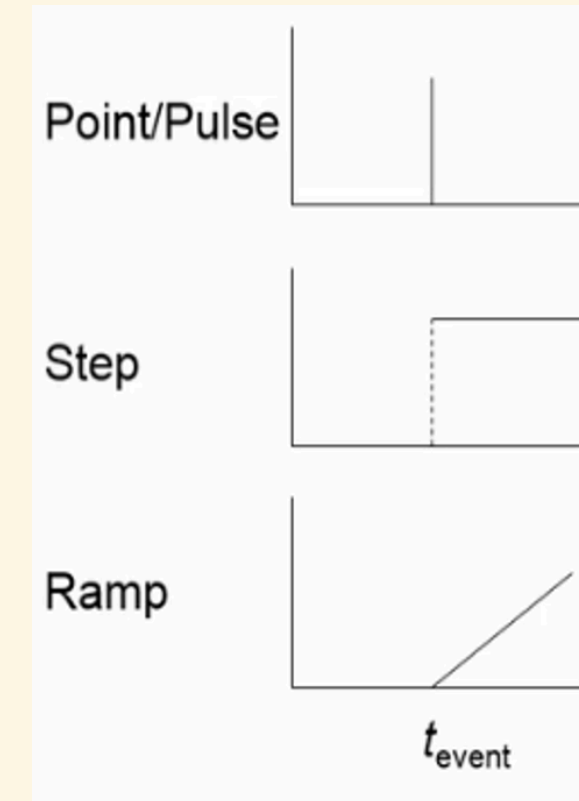
Intervention analysis: interventional ARIMA

$$y_t = \delta_0 I_t + w_t$$

- I_t is the intervention (see next slide)
- $w_t \sim \text{ARIMA}(p, d, q) \times (P, D, Q)$

Intervention analysis: interventional ARIMA

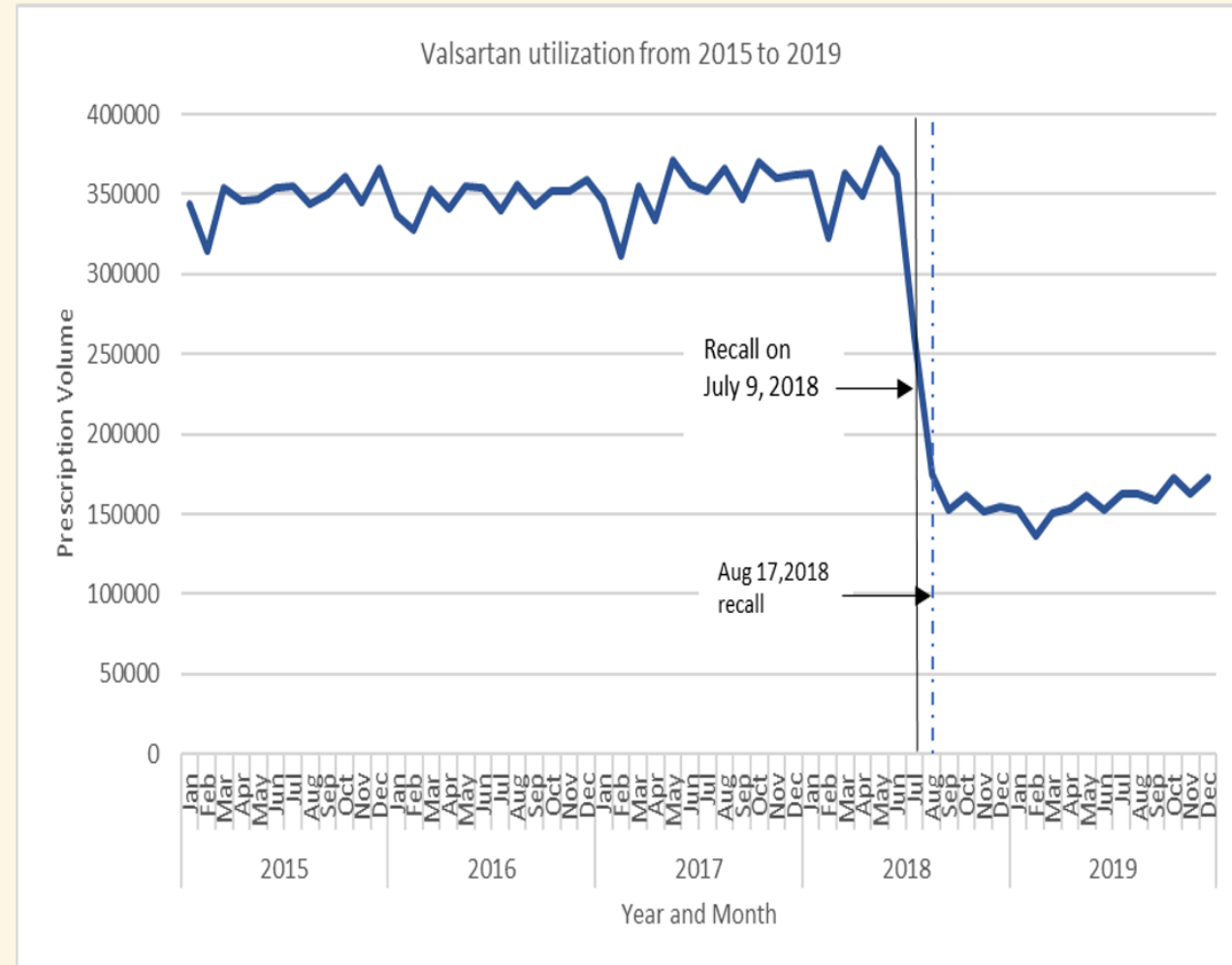
Intervention	Equation
Permanent step	$I_t = \begin{cases} 0, & t < T \\ 1, & t \geq T \end{cases}$
Temporary step over d	$I_t = \begin{cases} 0, & t < T \\ 1, & T \leq t < T + d \\ 0, & t \geq T + d \end{cases}$
Pulse	$I_t = \begin{cases} 0, & t \neq T \\ 1, & t = T \end{cases}$
Ramp	$I_t = \begin{cases} 0, & t < T \\ t - T + 1, & t \geq T \end{cases}$



- You can model multiple interventions (e.g., step + ramp)³
- Interventions should be pre-specified

Name that fit: #1

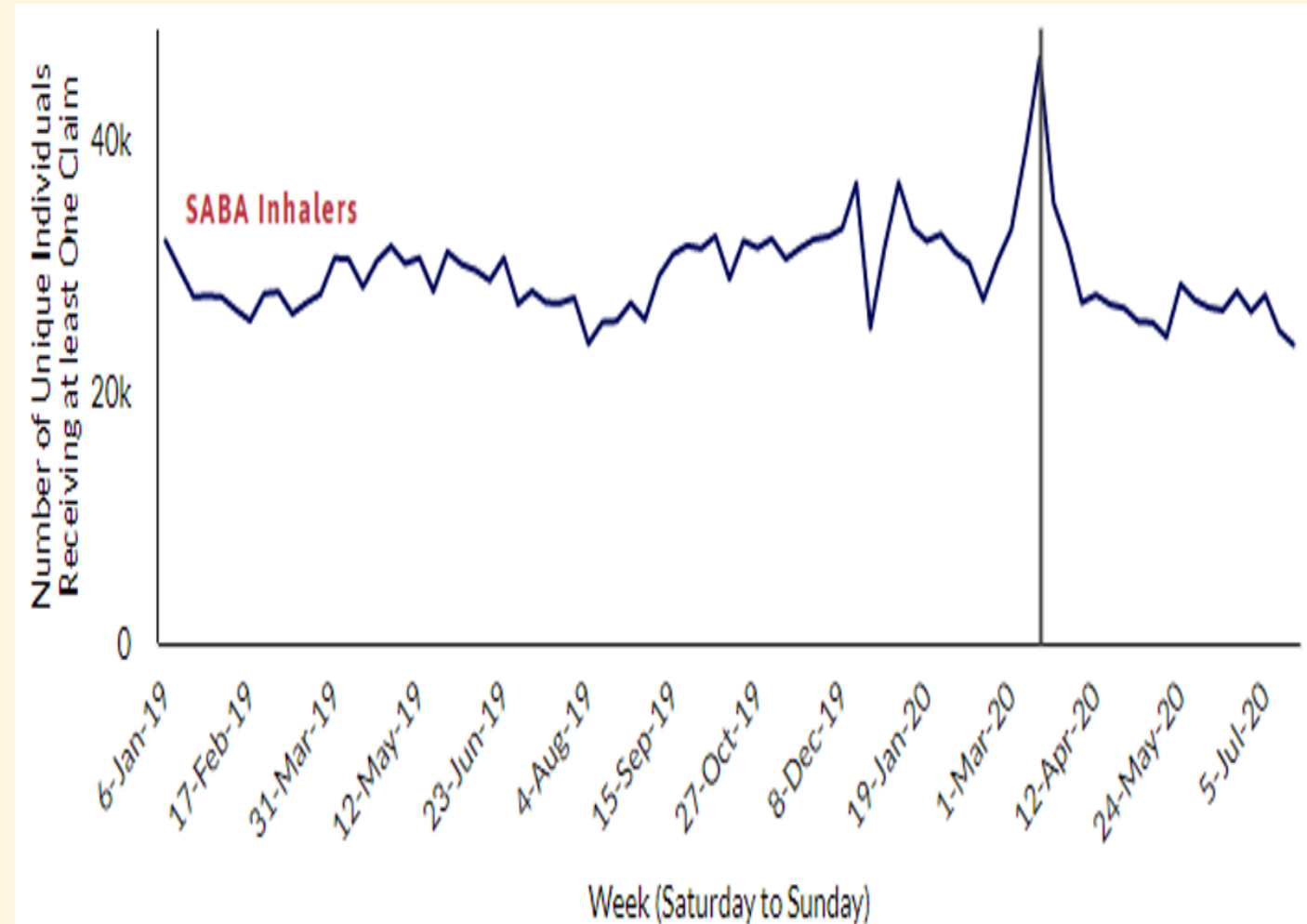
Scenario: Drug recall and shortage in July 2018



Name that fit: #2

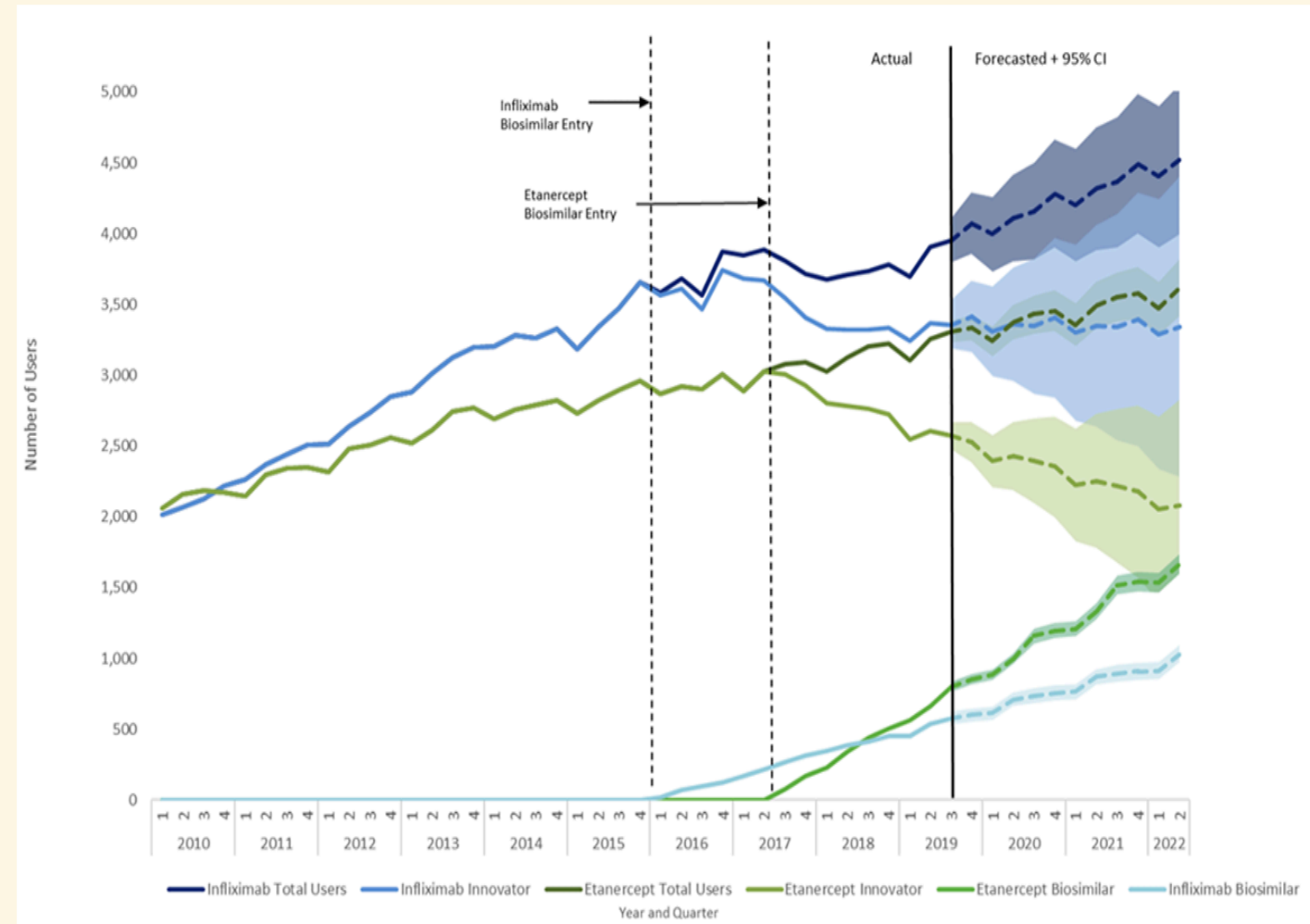
Scenario: COVID-19

pandemic lockdowns impact
on salbutamol (albuterol)
inhalers



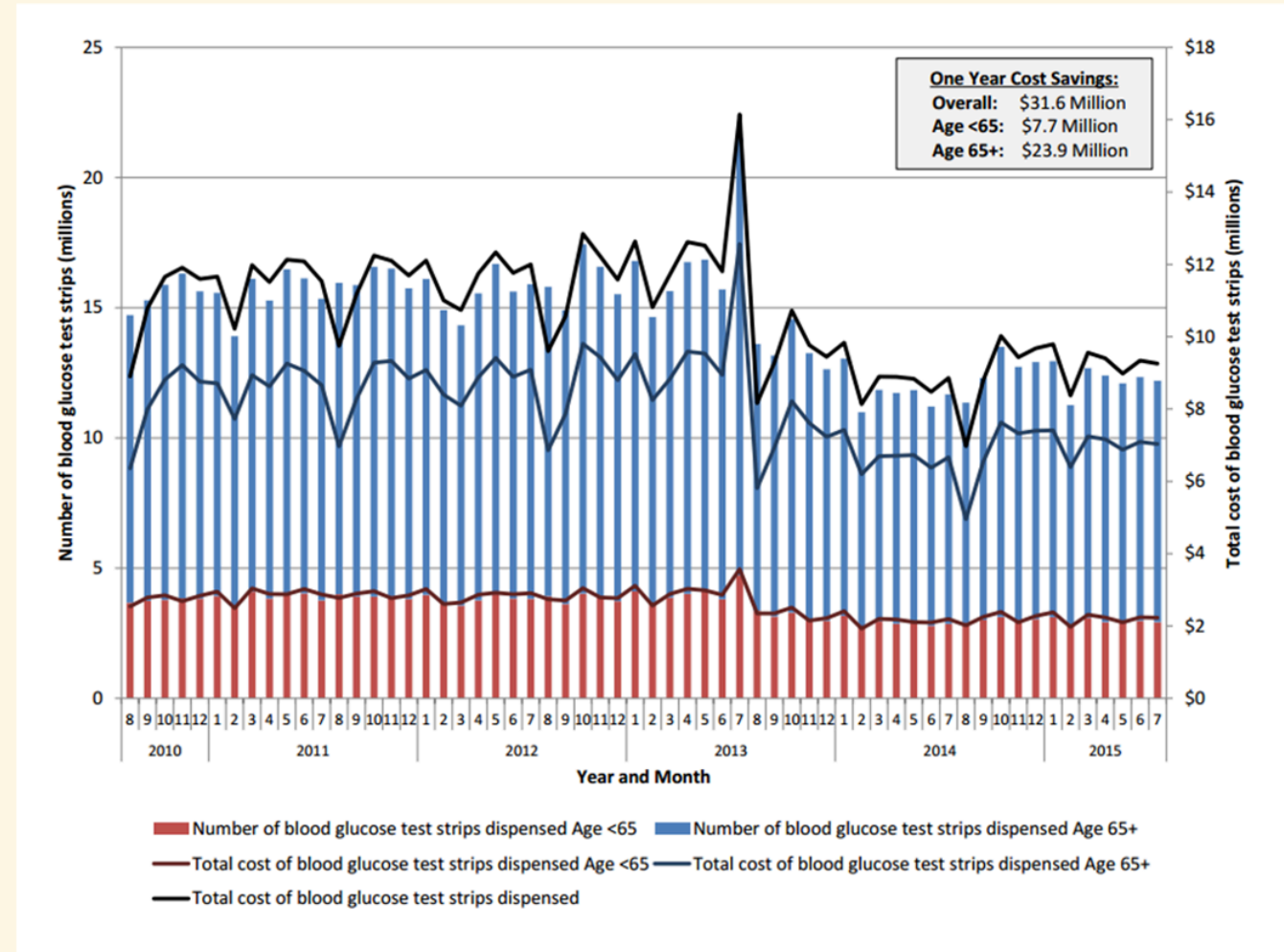
Name that fit: #3

Scenario: Introduction of a new biosimilar on formulary



Name that fit: #4

Scenario: New policy to limit access to a medication class, announced with one month until change



References

1. Gomes T, Martins D, Tadrous M, et al. Association of a Blood Glucose Test Strip Quantity-Limit Policy With Patient Outcomes: A Population-Based Study. *JAMA Intern Med*. 2017;177(1):61-66. doi:[10.1001/jamainternmed.2016.6851](https://doi.org/10.1001/jamainternmed.2016.6851)
2. PennState Eberly College of Science. STAT 510 Applied Time Series Analysis. Published online 2026. Accessed August 21, 2026. <https://online.stat.psu.edu/stat510/Lesson01>
3. Schaffer AL, Dobbins TA, Pearson SA. Interrupted time series analysis using autoregressive integrated moving average (ARIMA) models: A guide for evaluating large-scale health interventions. *BMC Med Res Methodol*. 2021;21(1):58. doi:[10.1186/s12874-021-01235-8](https://doi.org/10.1186/s12874-021-01235-8)